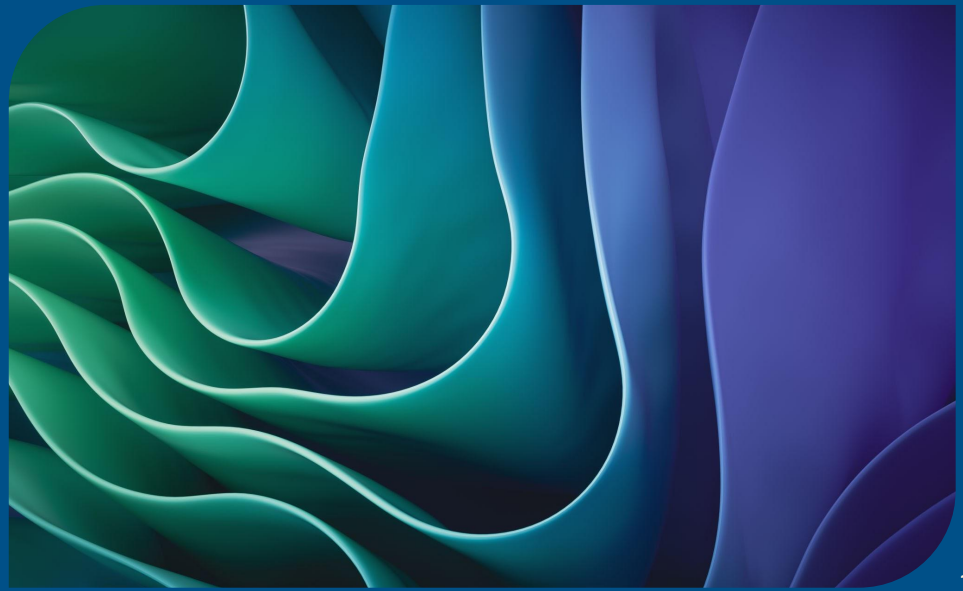
MIMIC-III Visualizations

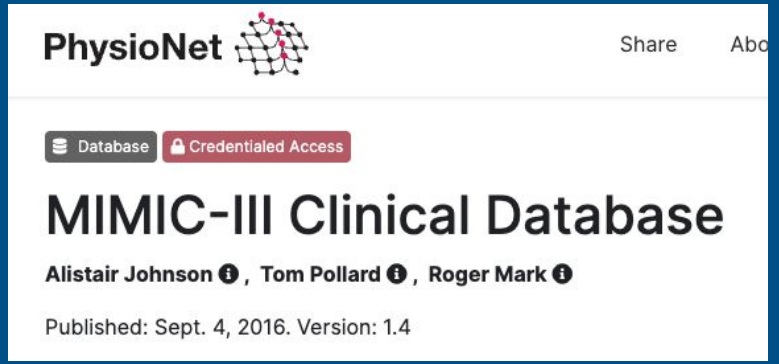
Shounak Ghosh
sg55893



1. Introduction
2. ECDF of Hospital Length of Stay
3. Hexbin Density: Age vs. ICU LOS
4. Parallel Coordinates: ICU Care Unit Profiles
5. Diagnosis Co-occurrence Network
6. Mosaic Plot: Admission Type vs Mortality
7. Bonus Dashboard: MIMIC-III Hospital and ICU Analytics Overview

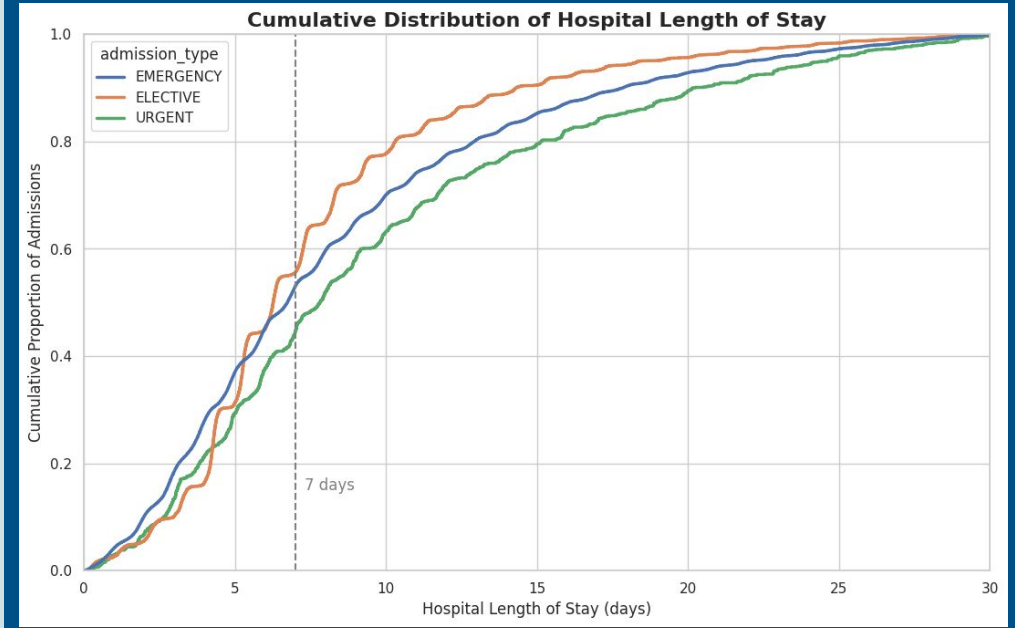
Introduction

- Access the MIMIC-III dataset with proper credentials and certifications
- Scroll to the bottom of the page linked above and download the ZIP file, or if it takes too long, the individual files listed below
- Unzip the folder and then upload the corresponding CSVs to the notebook cells (same style as the demo visualization notebook): PATIENTS.csv.gz, ADMISSIONS.csv.gz, DIAGNOSES_ICD.csv.gz, ICUSTAYS.csv.gz, D_ICD_DIAGNOSES.csv.gz
- Then, we perform some standardizations such as lowercase column names and standard datetimes
- We also build an admission-level analysis table by left-merging admissions with patients on subject_id. Calculate age at hospital admission from (admittime - dob) / 365.2425, and map de-identified ages above 89 to the 90+ group
- We also build an ICU-level analysis table by inner merging icustays with select admissions columns on subject_id, hadm_id
- These tables are created upfront to be used in the following 5 visualizations



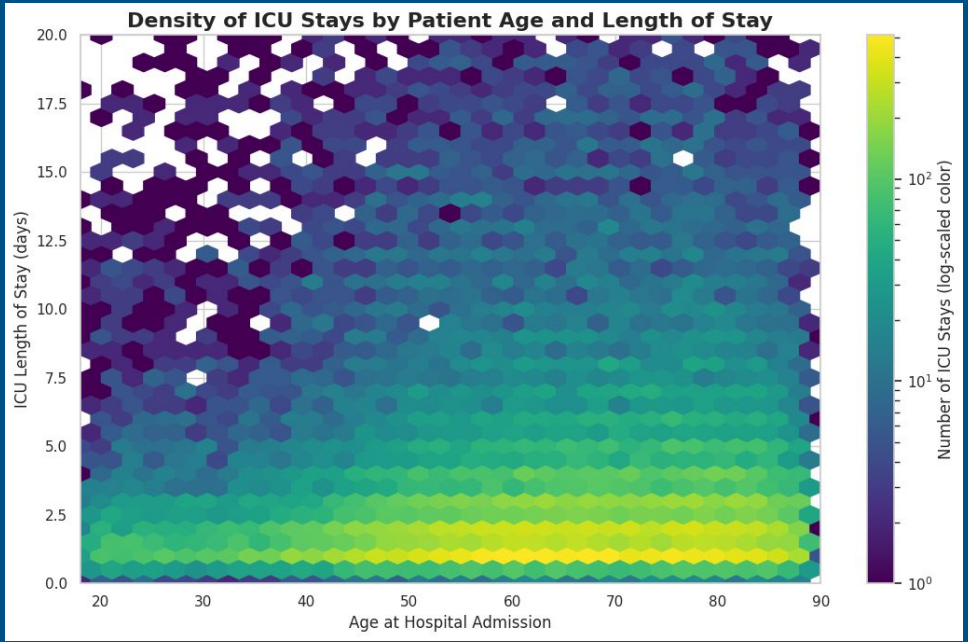
How Quickly do Hospital Stays End? An ECDF by Admission Type

- Background: An empirical cumulative distribution function (ECDF) answers the question “At a particular number of hospital days, what proportion of admissions have had a length of stay at or below that value?”
- Start with ADMISSIONS.csv.gz and PATIENTS.csv.gz
- Merge the tables using subject_id
- Keep adult patients and stays from 0-30 days to prevent extreme stays from dominating the graph
- Identify the 4 most frequent admission_type categories (only 3 present)
- Use sns.ecdfplot() with hospital LOS on the x-axis and admission type as the hue
- Add a vertical 7-day reference line
- Interpret the y-axis as the proportion of admissions having LOS less than or equal to a given x-value
- Curves that rise more quickly represent generally shorter hospital stays
- We see that approximately 45% of urgent visits, 52% of emergency visits, and 55% of elective visits have a hospital LOS less than or equal to a week



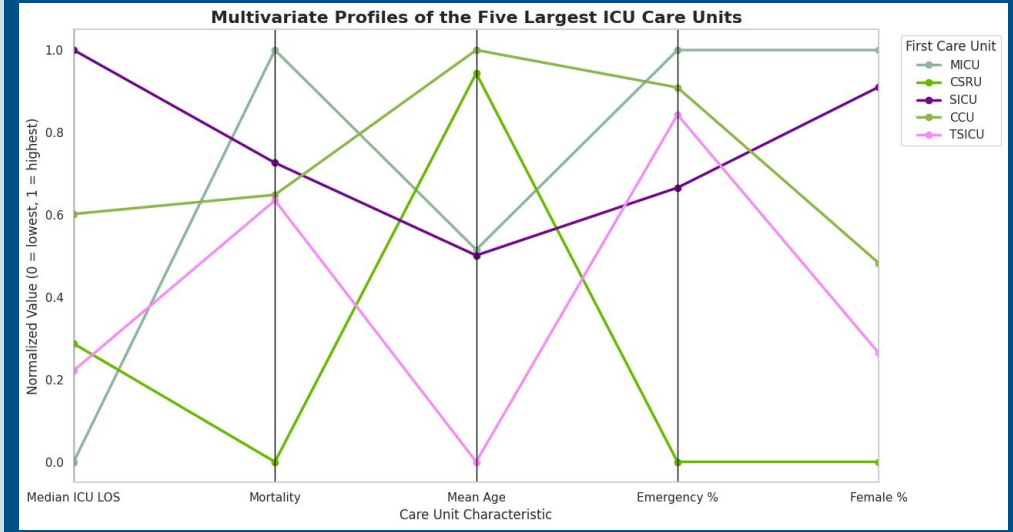
Where are ICU Patients Concentrated? Age vs. Length of Stay

- Background: Hexbin plots solve the overplotting problem that occurs when thousands of ICU observations are shown as individual scatter points
- We make use of the ICU-level analysis table created in the Introduction
- Restrict the visualization to adults and ICU stays of 0-20 days so extreme outliers do not compress the primary pattern
- Call plt.hexbin() instead of creating points for every ICU stay
- Set gridsize=35 to divide the graph into sufficiently detailed hexagonal regions
- Set mincnt=1 so empty areas remain blank
- Use bins="log" as some age/LOS combinations contain far more patients than others
- Add a color bar representing the number of ICU stays within each hexagonal region
- The highest-density region occurs among approximately ages 50-80 with ICU stays of roughly 1-3 days
- Longer ICU stays are much less common overall, as shown by the lower-density cells higher on the plot



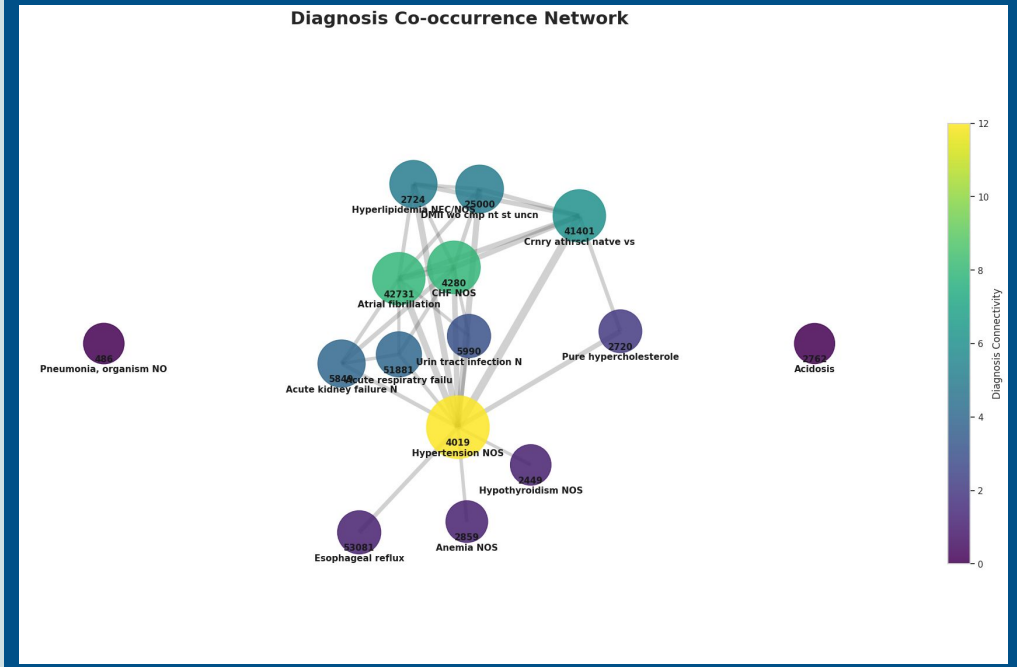
How Do ICU Care Units Differ? A Multivariate Profile

- Start with the ICU-level analysis table created in the Introduction and filter for only adults (age>=18)
- Group observations by first_careunit
- Calculate five care-unit characteristics:
  - Median ICU LOS
  - Hospital mortality rate
  - Mean patient age
  - Percentage of ICU stays linked to an Emergency hospital admission
  - Female share of ICU stays
- Count unique icustay_id values and keep the 5 largest care units so the graph remains readable
- Because LOS, age, mortality and percentages have different scales, normalize each characteristic using:
  - Normalized = (value - min)/(max - min)
  - Normalization means the graph shows relative differences between units, not raw numerical values
- Use parallel coordinates() from pandas.plotting
- Each vertical axis represents one metric
- Each line represents one ICU care unit
- MICU has an unusual profile with the highest normalized mortality and emergency-admission share but lowest normalized median ICU LOS



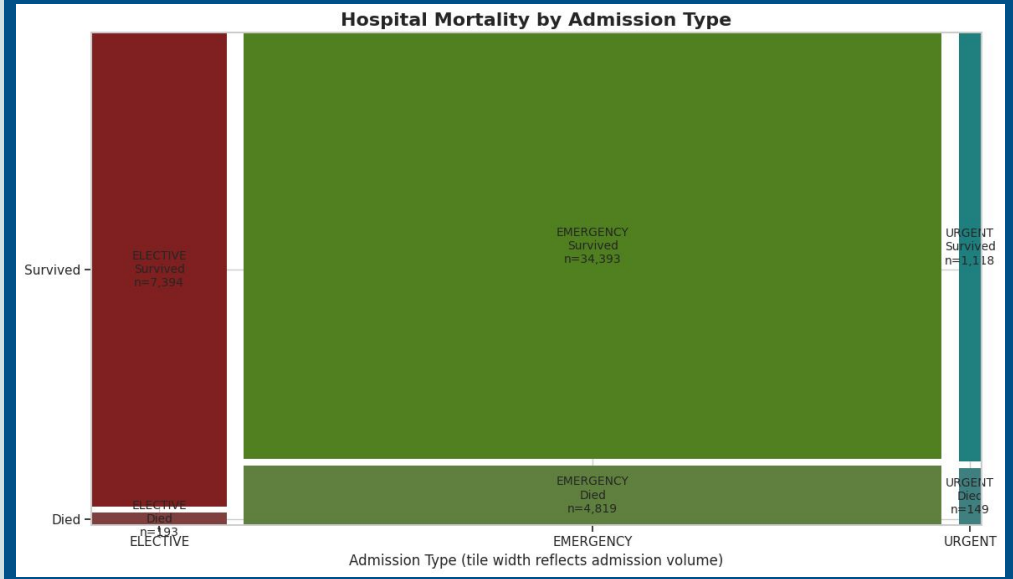
Which Diagnoses Travel Together? A Clinical Co-occurrence Network

- Start with DIAGNOSES_ICD.csv
- Restrict the analysis to adult hospital admissions
- Count the unique number of hadm_id values associated with each ICD-9 diagnosis
- Select the 15 most common diagnoses to prevent the graph from becoming unreadable
- For every hospital admission, create all possible pairs of diagnoses occurring during that admission
- Count how many admissions contain each pair
- Keep the 30 strongest diagnosis relationships
- Build a networkx.Graph()
    - Each node = ICD-9 diagnosis
    - Each edge = Diagnoses occurred together
- Set node size according to diagnosis prevalence
- Set edge thickness according to co-occurrence frequency
- Set the node color according to the number of other diagnoses connected to it
- Map ICD-9 codes to readable short_title labels using D_ICD_DIAGNOSES.csv.gz
- Use nx.kamada_kawai_layout() for the connected subgraph; scale the layout for readability and place degree-zero nodes separately at the edges
- Large nodes represent prevalent diagnoses; thicker edges represent stronger co-occurrence among the 30 retained pairs; node color represents connectivity within this filtered network



Does Hospital Mortality Differ by Admission Type?

- Start with the admission-level analysis table created in the Introduction
- Restrict the analysis to adult admissions
- Use hospital_expire_flag as the hospital mortality outcome
- Convert:
  - 0 → Survived
  - 1 → Died
- Select the 4 most common admission types (only 3 present)
- Count the admissions in every combination
- Pass these counts to statsmodels.graphics.mosaicplot.mosaic()
- Interpret tile width as the relative number of admissions in each admission category
- Interpret the vertical division within each category as its survival/death composition
- Because tile area represents frequency, the chart simultaneously communicates sample size and outcome differences
- Emergency and urgent admissions show substantially larger mortality shares (~12%) than elective admissions (~2.5%) in this adult cohort



Bonus Dashboard: MIMIC-III Hospital & ICU Analytics Overview

- Combine several complementary visuals so major hospital and ICU patterns can be compared at a glance
- Reuse the dataframes created for the individual visualizations
- Create a 2 x 2 dashboard using plt.subplots(2, 2, figsize=(18, 13))
- Panel A: Hospital LOS ECDF
  - Plot cumulative hospital LOS distributions by admission type
  - Add a 7-day reference line
  - Use this panel to compare how quickly different admission groups are discharged
- Panel B: Age vs. ICU LOS Hexbin
  - Plot patient age against ICU LOS
  - Aggregate observations into hexagonal cells instead of individual points
  - Use logarithmic color scale so both high and low density regions remain visible
- Panel C: ICU Care Unit Profiles
  - Show normalized metrics for the 5 largest ICU care units using parallel coordinates
  - Compare median LOS mortality, mean age, emergency admission percentage, and female percentage
- Panel D: Admission type vs. Mortality
  - Use a mosaic plot to compare survival and mortality across admission types
  - Tile width represents admission volume, while vertical division represents relative outcome composition
- Read panels together rather than independently for a compact view of patient flow, ICU utilization, care-unit characteristics, and mortality outcomes

